INTERNATIONAL TELECOMMUNICATION UNION

TELECOMMUNICATION STANDARDIZATION SECTOR

FOCUS GROUP ON AI NATIVE FOR TELECOMMUNICATION NETWORKS (FG-AINN)

FG-AINN-I-xxx

Original: English

Question(s): N/A New Delhi, 10th October, 2026

INPUT DOCUMENT

Source: Manipal Institute of Technology, Bengaluru

Title: Build-a-thon 5.0 2026: Autonomous Energy Aware Self Optimizing Radio Access Network (Auto EA-SORAN)

Contact:	Ms. Saanvi Chauhan Dept. of ECE, 4thYear Manipal Institute of Technology, Bengaluru	E-mail: saanvi.mitblr2023@learner.manipal.edu
Contact:	Ms. Nysa Suresh Sahu Dept. of ECE, 4th Year Manipal Institute of Technology, Bengaluru	E-mail: nysa.mitblr2024@learner.manipal.edu
Contact:	Mr. Anshul Katti Dept. of ECE, 4th Year Manipal Institute of Technology, Bengaluru	E-mail:anshul1.mitblr2024@learner.manipal.edu
Contact:	Dr. Ashutosh Anand Asst. Prof., Dept of ECE Manipal Institute of Technology	E-mail:ashutosh.anand@manipal.edu
Contact:	Mr. Rajat Duggal Senior System Specification Engineer MN RAN R&D, Nokia Bengaluru	E-mail: rajat.duggal@nokia.com

Abstract: This document describes a submission towards Build-a-thon 5.0, 2026 conducted in New Delhi on October 10th, 2026.

1. Introduction

1.a) Background to the Topic: The Energy Consumption Imperative

Although 5G New Radio (NR) is more energy-efficient per bit than 4G LTE, the overall energy consumption of 5G networks is expected to increase significantly. This is mainly due to network densification, massive Multiple-Input Multiple-Output (MIMO) antenna arrays, and the rapid growth in data traffic needed to meet 5G's performance requirements.

These factors pose major operational and environmental challenges. The Radio Access Network (RAN) consumes around 75–85% of a mobile network's total energy, making it a significant operational expenditure (OPEX) and affecting financial sustainability. Additionally, operators face increasing pressure to achieve sustainability targets, including the UN's Sustainable Development Goal 13 (Climate Action). One major source of inefficiency is the always-on operation of base stations, which continue to consume substantial power even during periods of low traffic. This highlights the gap between static resource allocation and dynamically changing network demand.

1.b) Need for AI-Native Network Automation

The complexity of 5G RANs makes traditional rule-based Network Management Systems (NMS) inadequate, as they are often static, manual, and reactive to changing network conditions. This creates a need for AI-native networks that integrate intelligence into the network infrastructure, enabling autonomous and proactive control through distributed, agent-based architectures. Such systems establish a closed-loop process of observation, prediction, decision-making, and actuation.
AI-native automation is also crucial for supporting network slicing, a key feature of 5G that enables services such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC) to operate on shared infrastructure. Each service has different energy and performance requirements. Static energy-saving policies may compromise Service Level Agreements (SLAs) or miss opportunities for optimization. Therefore, an intelligent system that dynamically balances energy efficiency and Quality of Service (QoS) for each service is essential.

.

2. State of the Art in 5G Energy Management

2.a) Commercial AI-Powered Platforms

Major telecom vendors are incorporating AI-driven energy management into their 5G RAN solutions, shifting from static configurations to dynamic, data-driven optimization.

- One vendor uses an ML-based engine to dynamically adjust cell sleep-mode thresholds, reporting energy savings of 10–12%.
- Another uses the MantaRay Self-Organizing Network (SON) platform to enable closed-loop automation, claiming up to 30% additional power reduction.
- A third applies AI-based traffic prediction to automate power amplifier control and multi-level radio sleep modes.

Although these solutions demonstrate the growing adoption of intelligent network automation, they are often proprietary, potentially limiting transparency, flexibility, and the ability to integrate more advanced AI-native frameworks.

2.b) Academic Research Contributions

Academic research has applied machine learning to 5G energy efficiency across several key areas. Foundational work includes **traffic prediction and load forecasting** using models like LSTMs to enable proactive control. Building on this, a dominant theme is **dynamic base station (BS) on/off switching**, using techniques from heuristics to Reinforcement Learning (RL) to manage sleep modes. Other significant areas include the use of **Multi-Agent Systems (MAS)** for robust, distributed control ; ML-based optimization

of **resource allocation** ; and **Federated Learning (FL)** for privacy-preserving model training.

2.c) Standardisation Efforts in Energy Efficiency

Standardization work by 3GPP's Working Group 5 (SA5) provides the foundation for scalable network automation, defining standards for Energy Efficiency (EE) and Energy Saving (ES) across the entire 5G system. A key contribution is the formal definition of the EE KPI:

$$EE = \frac{Performace}{Energy\ Consupmtion\ (EC)}$$

This ratio links energy consumption to network performance, where “Performance” varies according to the service type—for example, data throughput for eMBB and reliability for URLLC. 3GPP also defines standardized data collection methods through Operations, Administration, and Maintenance (OA&M) interfaces. These standards provide a vendor-neutral way to measure energy efficiency (EE), offering a reliable and interoperable target for AI optimization algorithms. Therefore, using 3GPP-defined Key Performance Indicators (KPIs) is an important design choice.

3. Research Gaps and System Limitations

Analysis of the state of the art reveals key gaps that the Auto EA-SORAN project addresses.

- **Holistic vs. Siloed Designs:** Many existing solutions optimize individual parameters independently, such as cell sleep modes, without considering the interdependencies between different control mechanisms. The proposed multi-agent architecture addresses this limitation by coordinating multiple actions, such as transmit power adjustment and sleep scheduling, to achieve more holistic and effective network energy optimization.
- **The High-Fidelity Validation Gap:** Much of the existing academic research relies on simplified simulators that fail to capture real-world protocol stack interactions, such as how reducing transmission power can increase MAC-layer retransmissions. This project uses Simu5G as a high-fidelity digital twin to model the 5G network stack more realistically, helping ensure that control policies remain robust under varying network conditions. It also addresses the data gap by generating realistic synthetic data for training AI models.
- **The Explainability Deficit:** The black-box nature of many AI models can reduce operator trust. This project addresses this through architectural explainability. The Feedback Agent acts as a transparent validation mechanism, checking every action against clear, 3GPP-compliant performance thresholds. Rather than relying on a complex model, it uses understandable logic to validate decisions, making the system's behaviour more transparent and trustworthy by design.
- **Static Management of the QoS-Energy Trade-off:** Existing systems often rely on fixed, network-wide thresholds to balance energy savings and Quality of Service (QoS), making them unsuitable for the diverse requirements of 5G services. The proposed system dynamically manages this trade-off at the service level. The Feedback Agent applies different validation policies for each traffic type, helping maximize energy savings while ensuring Service Level Agreements (SLAs) are maintained.

4. Technological Foundations and Proposed Architecture (Auto EA-SORAN)

4.a) A Closed-Loop Multi-Agent Control System

The Auto EA-SORAN system is a closed-loop control architecture composed of four cooperative agents executing a continuous cycle of monitoring, prediction, optimization, and actuation.

- **Prediction Agent:** Ingests real-time network state data and uses an ML model to forecast future energy consumption.
- **Optimization Agent:** Receives the forecast and formulates a multi-faceted energy conservation plan (e.g., adjust power, enter sleep mode).
- **Feedback Agent:** Validates the proposed plan against predefined policies and 3GPP-compliant performance thresholds to prevent QoS degradation.
- **Orchestration Agent:** Executes the final, validated plan within the simulation environment.

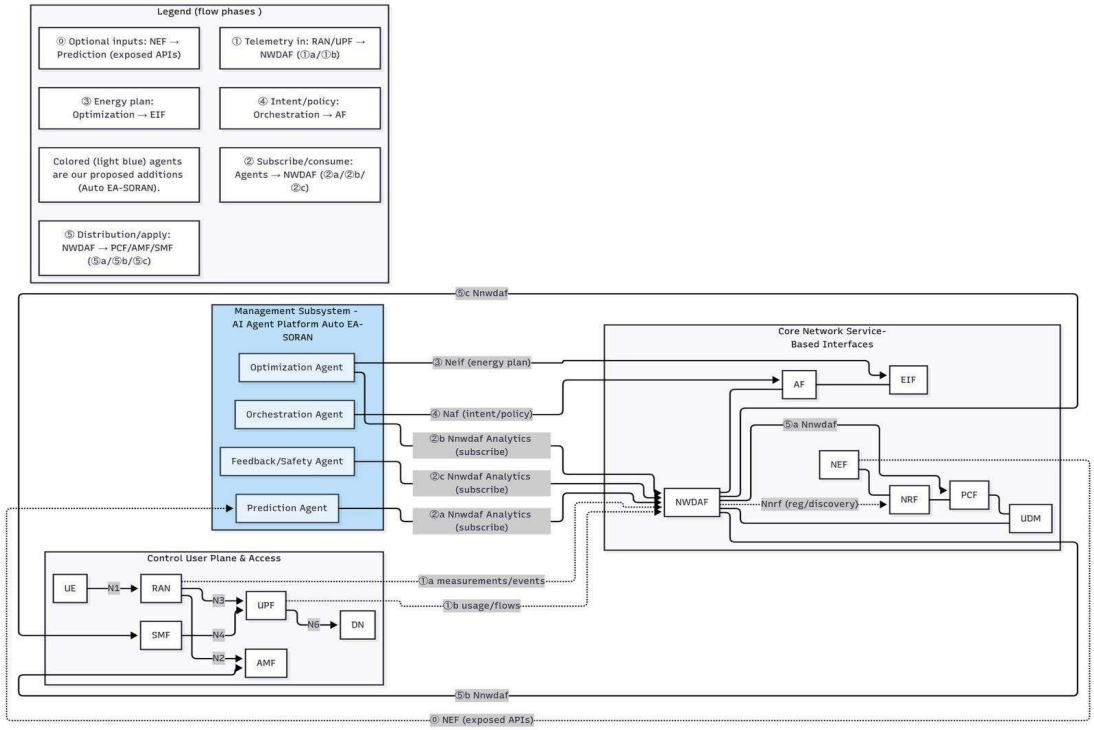
4.b) Simu5G as a High-Fidelity Digital Twin

The system is developed and validated using the Simu5G network simulator, which serves as a high-fidelity digital twin of the 5G RAN and models the NR protocol stack. A key innovation is a custom, state-based gNB energy model developed by extending INET's Energy Framework. It defines discrete power states, such as active, micro-sleep, and deep sleep, while modeling active power consumption as a linear function of real-time Physical Resource Block (PRB) utilization.

$$Ptotal = Pbase + Ptraffic \times \left(\frac{num_allocated_PRBs}{total_available_PRBs} \right)$$

This ensures the model is realistically coupled with data plane activity. To maintain a responsive control loop, the architecture uses a low-latency, socket-based Inter-Process Communication (IPC) mechanism (ZeroMQ), between the Python agents and the C++ simulation.

4.c) Proposed Architecture mapped to 3GPP:



5. Current Status & Demo Proposal

Our project, “Autonomous Energy Aware Self Optimizing RAN,” will culminate in a demonstration of a complete, closed-loop optimization cycle within the simulated 5G network. The demonstration will visualize the following sequence:

- The system observes the network state in Simu5G (e.g., PRB utilization, active user count).
- The Prediction Agent forecasts future traffic and energy consumption.
- The Optimization Agent proposes a specific energy-saving action (e.g., "transition gNB-3 into state_micro_sleep").
- The Feedback Agent validates that this action will not violate the predefined QoS policies for active users.
- The Orchestration Agent successfully transmits and executes the command within the Simu5G environment.
- The results will be shown via the simulation's output vectors and scalars, illustrating a measurable decrease in the target gNB's power draw while maintaining stable user throughput. A comparative analysis against a baseline (unoptimized) simulation run will be presented to quantitatively demonstrate the percentage of energy saved.

Submission Id	FG-AINN-PoC-xxx (Id number is to be assigned by the secretariat)
PoC Objective	In this Build-a-thon proposal, we aim to study **AI-native, closed-loop energy optimization for 5G base stations** using a prediction → optimization → feedback→ actuation cycle inside **Simu5G**. **Specifically,** we will demonstrate: 1. a **Prediction Agent** that forecasts short-horizon traffic and energy, 2. an **Optimization Agent** that selects an energy-saving action (e.g., TX-power trim, carrier off, micro/deep sleep) under **QoS guard-rails**, 3. a **Feedback/Safety Agent** that validates and can **roll back** actions, and 4. an **Orchestration Agent** that executes the command in Simu5G via a low-latency socket. **At this point, we hypothesize that** this closed loop will deliver **≥15–25% energy reduction** on targeted gNBs **without violating QoS** (≤3% p95 throughput degradation; latency/loss within policy).
PoC Setup	In this proposal we use the **Farzi project’s three CSV datasets** (static, offline) **plus real-time telemetry from Simu5G**. The three CSV files train the energy-prediction model; **Simu5G** feeds the trained model with live network metrics (e.g., PRB utilization, active UEs, throughput/latency) during the demo. **Model training is done using** Python (**scikit-learn/Keras**) with engineered temporal features; **SHAP** provides model-level explainability. The agents communicate with an in-simulation **Orchestration Agent** over **ZeroMQ/TCP**; energy is measured via a **state-based gNB consumer** (active / micro-sleep / deep-sleep) that depends on PRB utilization.
PoC demo and Succ	The following scenarios would be demonstrated: 1) **Closed-loop saving:** the system predicts a low-load window and executes an action (e.g., “gNB-3 → micro_sleep” or “ΔP = −2 dB”); vectors/scalars show **lower power** while UE throughput remains within policy.

ess Criteria	2) **Intent-to-policy orchestration:** an operator intent such as "**minimize energy with p95 UE throughput ≥ X and latency ≤ Y**" is translated to policies; when conditions are met, an energy action is applied and logged with an **explanation trace**. 3) **Safety & rollback:** if post-actuation KPIs breach guard-rails, the **Feedback Agent** auto-reverts to the last safe configuration and records the event.
PoC Plan and Report	Stage-1: Demo proposal by September 07, 2025 Stage-2: Implementation proposal and test cases by September 15, 2025 Stage-3: Initial implementation by September 30, 2025 Stage-4: Final demo and report in FG AINN template by October 9, 2025
Created by	Team: NoWiresAttached Mentor: Rajat Duggal Team lead: Ms. Saanvi C. Team members: 1. Mr. Anshul Katti 2. Ms. Nysa Sahu Internal guide: Asst. Prof. Dr. Ashutosh Anand
Creation date	23 October 2026
Open problems and future study	While our current proof-of-concept demonstrates prediction and autonomous self-optimization through explainable AI and lightweight agents, several open challenges remain for further research and deployment at scale: 1. **Multi-cell coordination:** energy savings with HO/load-balancing awareness across clusters. 2. **Real-time constraints:** tighter inference/optimization deadlines and jitter control on commodity edge hardware. 3. **Policy richness:** extend guard-rails to include SLA tiers, slice-specific constraints, and time-of-day pricing signals. 4. **Model drift & continual learning:** handle seasonality and traffic shifts; evaluate **federated**/online learning. 5. **Wider actuator set:** PA bias control, beam/PRB budgeting, scheduler knobs; safety proofs for each. 6. **Standards alignment:** map to **O-RAN** E2 control (near-RT RIC xApps) for field portability. 7. **Adversarial robustness:** spoofed telemetry/conflicting intents; formal verification of rollback logic. These open problems provide an opportunity for iterative enhancement of our framework toward a production-ready, standard-compliant AI-Native self-optimizing system suitable for the energy conservation of 5G Base Stations.
References	[1] "A Calibrated and Automated Simulator for Innovations in 5G," *arXiv*, 2024. https://arxiv.org/html/2404.10643v1 [2] "Scalable Real-Time Emulation of 5G Networks With Simu5G," *ResearchGate*, n.d. https://www.researchgate.net/publication/355690617

[3] "Detailed Explanation on 5G Simulation: Parameters • Model • Indicators," *OMNeT++ Tutorial*, n.d. https://omnet-tutorial.com/5g-simulation/

[4] "Scalable Real-Time Emulation of 5G Networks With Simu5G," *SciSpace (PDF)*, n.d. https://scispace.com/pdf/scalable-real-time-emulation-of-5g-networks-with-simu5g-2mv2rbff4t.pdf

[5] "5G-Energy-Consumption-Modelling-Solution — Team Farzi Data Scientists," *GitHub (ITU AI/ML in 5G Challenge)*, n.d. https://github.com/ITU-AI-ML-in-5G-Challenge/5G-Energy-Consumption-Modelling-Solution-Team-Farzi-Data-Scientists

[6] "How to Implement 5G Network Simulator Projects (Guide)," *Network Simulation Tools*, n.d. https://networksimulationtools.com/5g-network-simulator/

[7] "A Survey on Green Enablers: A Study on the Energy Efficiency of AI-Based 5G Networks," *MDPI (Sensors)*, 2024. https://www.mdpi.com/1424-8220/24/14/4609

[8] "Overview — Simu5G: Simulator for 5G New Radio Networks," *Simu5G Users' Guide*, n.d. https://simu5g.org/users-guide/overview

[9] "Running Simu5G as a Real-Time Network Emulator," *Simu5G Users' Guide*, n.d. https://simu5g.org/users-guide/emulation

[10] "Simu5G: A System-level Simulator for 5G Networks," *SciTePress (PDF)*, 2020. https://www.scitepress.org/Papers/2020/98264/98264.pdf

[11] "Simu5G: Simulator for 5G New Radio Networks," *Project site*, n.d. https://simu5g.org/

[12] "Modeling Power Consumption," *INET 4.5.4 Documentation*, n.d. https://inet.omnetpp.org/docs/users-guide/ch-power.html

[13] L. M. Feeney, "An Extensible Framework for Simulating Battery Consumption in Wireless Networks," n.d. https://sonri.org/publications/Files/omnet10energy.pdf

[14] "How to Calculate Network Energy Utilization in OMNeT++," *OMNeT++ Manual (community)*, n.d. https://omnet-manual.com/how-to-calculate-network-energy-utilization-in-omnet/

[15] "AI-based Energy Consumption Modeling of 5G Base Stations: An Energy-Efficient Approach," *figshare (conference contribution)*, n.d. https://sussex.figshare.com/articles/conference_contribution/AI-based_energy_consumption_modeling_of_5G_base_stations_an_energy_efficient_approach/26394829

[16] "Modeling of IoT Devices Energy Consumption in 5G Networks," *mediaTUM (PDF)*, n.d. https://mediatum.ub.tum.de/doc/1705687/55ahjo0udp2jr6z1159ilkwbr.Modelling_energy_consumption_of_IoT_devices_in_5G_network.pdf

[17] "Energy Efficiency for 5G and Beyond 5G: Potential, Limitations, and Future Directions," *MDPI (Sensors)*, 2024. https://www.mdpi.com/1424-8220/24/22/7402

[18] "Energy Consumption Modelling for 5G Radio Base Stations with Machine Learning," *Lund University Publications (PDF)*, n.d. https://lup.lub.lu.se/student-papers/record/9133984/file/9133985.pdf

[19] "BaseStationStatsCollector," *Simu5G NED Documentation*, n.d. https://simu5g.org/neddoc/simu5g.corenetwork.statsCollector.BaseStationStatsCollector.html

[20] "Packages — Simu5G," *Simu5G NED Documentation*, n.d. https://simu5g.org/neddoc/packages.html

[21] "OMNeT++ 5G Thesis Ideas," *PHD Services*, n.d. https://phdservices.org/omnet-5g/

[22] "New Radio Tutorial — Simu5G," *Tutorial docs*, n.d. https://simu5g.org/tutorials/NR/doc/index

[23] "BackgroundScheduler," *Simu5G NED Documentation*, n.d. https://simu5g.org/neddoc/simu5g.nodes.backgroundCell.BackgroundScheduler.html

[24] "Simu5G — 5G NR and LTE/LTE-A User-Plane Simulation Model for OMNeT++ & INET," *GitHub*, n.d. https://github.com/Unipisa/Simu5G

[25] "A Network Simulator-based Method of Dynamic Traffic Generation for 5G Network Slicing," *Concordia University — Spectrum*, n.d. https://spectrum.library.concordia.ca/id/eprint/990258/1/Ding_MCompSc_df

[26] "Parameter Studies," *OMNeT++ Technical Articles — TicToc Tutorial*, n.d. https://docs.omnetpp.org/tutorials/tictoc/part7/

[27] "gNodeB," *Simu5G NED Documentation*, n.d. https://simu5g.org/neddoc/simu5g.nodes.NR.gNodeB.html

[28] "OMNeT++ — Simulation Manual," *Omnetpp.org*, 2025. https://doc.omnetpp.org/omnetpp5/manual/

[29] "OMNeT++ Simulation Library: cSimpleModule Class Reference," *Omnetpp.org (API v4)*, n.d. https://doc.omnetpp.org/omnetpp4/api/classcSimpleModule.html

[30] "Simu5G: A System-level Simulator for 5G Networks," *ARPI (PDF)*, https://arpi.unipi.it/retrieve/handle/11568/1039566/609208/SIMULTECH_2020.pdf

[31] "Simu5G – An OMNeT++ Library for End-to-End Performance Evaluation of 5G Networks," *ARPI (PDF)*, 2020. https://arpi.unipi.it/bitstream/11568/1053174/2/2020%20Access.pdf

[32] "Steps to Implement Network Energy Management in OMNeT++," *OMNeT++ Manual (community)*, n.d. https://omnet-manual.com/how-to-implement-network-energy-management-in-omnet/

[33] "Energy Efficiency (EE) SA5 Work and Results," *3GPP*, n.d. https://www.3gpp.org/technologies/energy-efficiency-ee-sa5-work-and-results

[34] "5G and Energy Efficiency," *5G-PPP Standards Tracker (booklet)*, n.d. https://standards-tracker.5g-ppp.eu/sites/default/files/BookletA4_EnergyEfficiency.pdf

[35] "A Comparison Analysis of 5G Key Performance Indicators Based on Entropy," *ResearchGate (PDF)*, n.d. https://www.researchgate.net/publication/386292921

[36] "Main Modules of the Simu5G Model Library (figure)," *ResearchGate — Scientific Diagram*, n.d.

	https://www.researchgate.net/figure/Main-modules-of-the-Simu5G-model-library_fig3344434106 [37] "SDAP-based QoS Flow Multiplexing Support in Simu5G for 5G NR Simulation," *arXiv*, 2025. https://arxiv.org/html/2508.12785v1

6. Implementation proposal (Build-a-thon Stage-2 proposal)

This clause describes the implementation proposal for the submission above. Implementation is done in the following phases. Phase-1 includes preparation and demo of the test setup. Phase-2 includes a demo of (opensource) base code in as-is-where-is condition. Phase-3 includes modification of the base code and test setup to suit the requirements of the demo mentioned in clause-1. Phase-4 includes a documentation of results corresponding to the test cases executed.

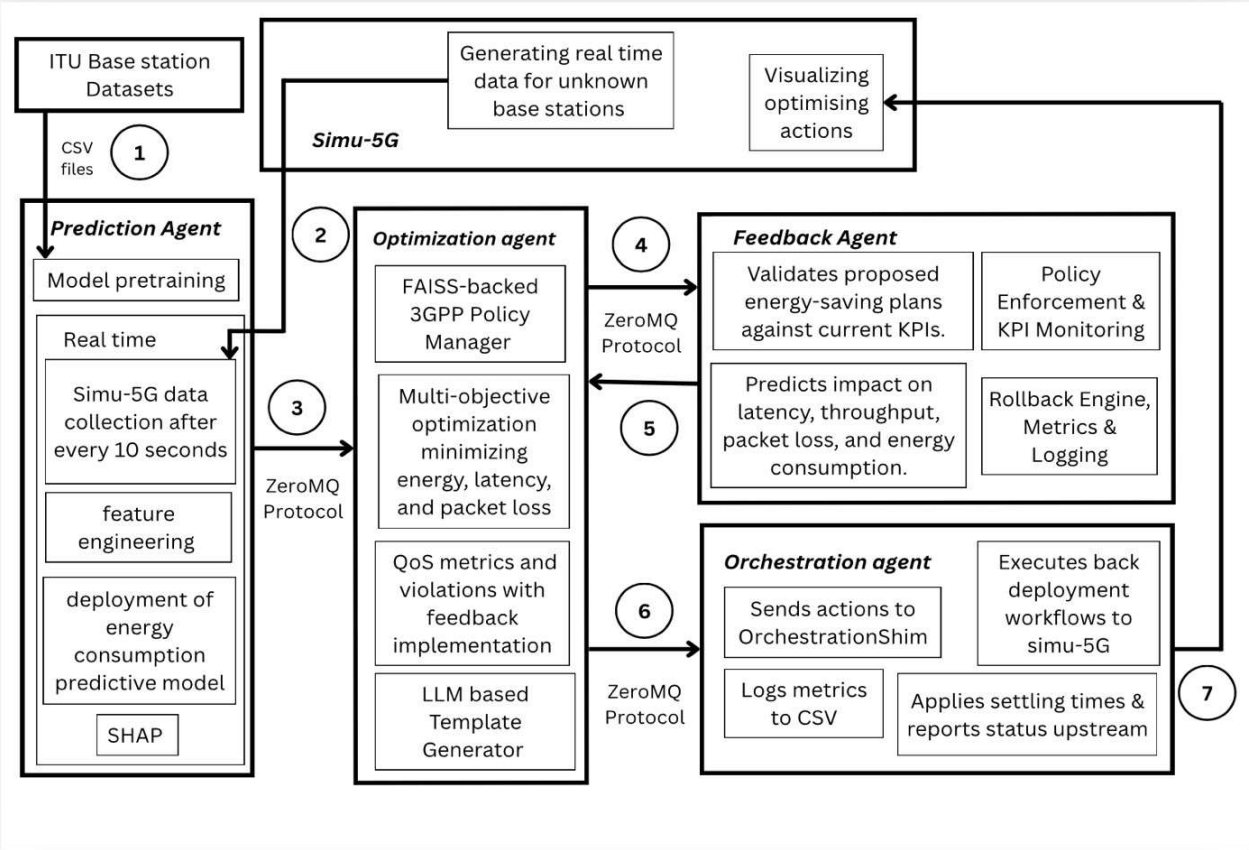
6.a) Description of the test setup

The demo instantiates an AI-native, closed-loop control system integrated with **Simu5G**. Four agents—**Prediction**, **Optimization**, **Feedback**, and **Orchestration**—exchange messages over a **ZeroMQ** socket with an in-simulation orchestration proxy to apply actions on the gNB at runtime. Simu5G serves as a high-fidelity **digital twin** of the 5G RAN; energy is modelled via discrete states (active / micro-sleep / deep-sleep) with active power tied to **PRB utilization**.

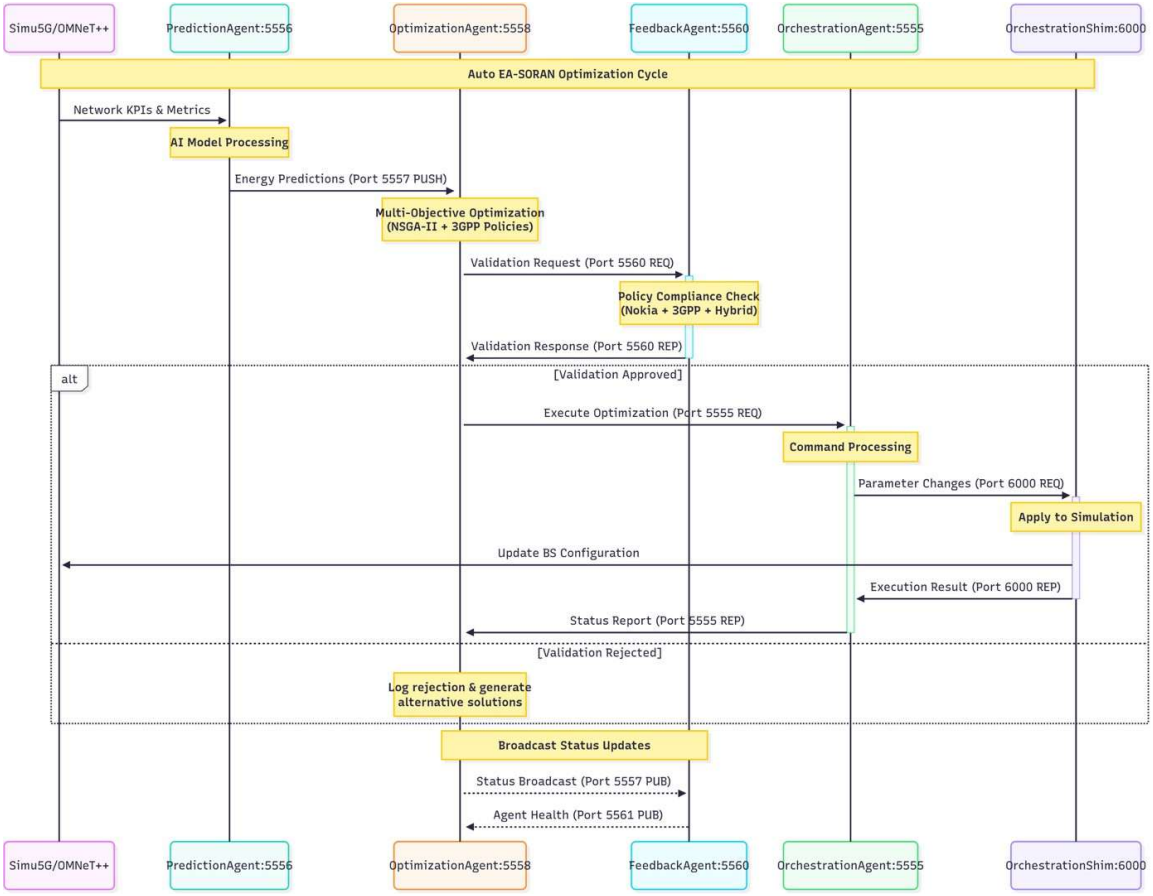
The test setup involves the following:

- **Code generation model from:**
  OpenAI API (LLM). The Optimization Agent uses an LLM to draft structured **plan objects** (e.g., *ΔP = −2 dB on gNB-k*, *enter micro_sleep*, *deactivate one CC*), which are then checked against policy guard-rails by the Feedback Agent before actuation.
- **Service orchestrator from:**
  In-simulation Orchestration Agent (C++ in Simu5G) with a thin Python client over ZeroMQ.
  The Python side sends validated actions; the C++ **Orchestration Agent** receives, parses, and applies them by updating gNB parameters or triggering state transitions during the run (e.g., `txPower`, sleep-mode entry, carrier on/off).
- **Agent framework from:**
  *Lightweight Python services (Prediction → Optimization → Feedback → Orchestration) communicating via **ZeroMQ/TCP**.*
  A brokerless, asynchronous IPC ensures low-latency state→decision→actuation, avoiding stale decisions typical with file-based exchange.
- **Simulator from:**
  Simu5G (OMNeT++/INET).
  Simu5G acts as the controllable digital twin. We couple a **state-based gNB energy model** to real-time PRB activity so that power reflects load; vectors/scalars are recorded for post-run analysis.
- **Datasets from:**
  Farzi project CSVs (offline training) + live telemetry from Simu5G (runtime).
  Three CSV files train the Prediction Agent; at demo time, Simu5G streams **PRB utilization, active UEs, and throughput/latency** as features to the agents. Model training uses Python with engineered temporal features; **SHAP** provides explainability.

The following figure shows the pipeline of control in the test setup:-



The sequence diagram for the above pipeline is as follows:-



6.b) Description & reference to base code

The base code is used from:
GitHub (Team Farzi): ITU-AI-ML-in-5G-Challenge / 5G-Energy-Consumption-Modelling-Solution-Team-Farzi-Data-Scientists — final Keras ANN with feature engineering, CV strategy, and SHAP analysis.

The complete code for Autonomous Energy Aware Self Optimizing RAN, built on top of their work can be found on our GitHub (link)

It has the following components:

- **Keras ANN final model** (tabular regression) with embeddings for categorical features and dense layers for continuous features; SiLU/Swish activation; trained with MAE/MAPE/WMAPE focus.
- **Feature engineering pipeline:** time-stamp features (day/weekday/hour + periodic spline basis), lagged features (T-1/T-2/T-3 for load/ESModes/Energy), binned load, and signal-processed load (Savitzky–Golay and SOS Butterworth filters) — strictly **no future data**.
- **Cross-validation:** GroupKFold over base-station IDs to enforce generalization to unseen BS/products/configs.
- **Explainability:** SHAP analyses for existing vs new base stations.
- **Reproducible training environment** (Kaggle GPU; pinned libs/versions; notebooks and submission artefacts).

Of these, the following are important for the demo:

- The **Keras ANN** predictor (selected as final in Farzi's solution due to best cross-val and leaderboard performance) to estimate gNB energy from live features.
- The **feature set definition** (hourly temporal features, lags, load transforms) that we map to Simu5G telemetry for real-time inference.
- The **no-future-values discipline** to keep the closed loop causal.
- **SHAP** for quick sanity checks on actionability during dry-runs.

The following are used as is:

- **Model architecture & loss/metrics:** Keras ANN with embedding + dense stacks; MAE/MAPE tracking; WMAPE context retained for comparability.
- **Core feature engineering logic:** hour/weekday/periodic splines; lagged features; binned load; SG/SOS filters (implemented exactly as in the base write-up).
- **GroupKFold** generalization approach for any offline re-training runs.

The following are changed as below:

- **I/O & runtime integration:** Replaced file/CSV exchange from the notebooks with **ZeroMQ** message passing between **Prediction → Optimization → Feedback → Orchestration** agents and Simu5G; the predictor now runs as a long-lived Python service. *(Reason: low-latency, brokerless IPC for closed-loop control.)*
- **Feature availability mapping:** Limited the online feature set to those emitted by Simu5G at 1-hour/short-horizon cadence (e.g., PRB utilization, active UEs, ES mode counters, config params). Engineered features (splines, lags, filters) are computed **on the fly** inside the Prediction Agent from these inputs. *(Reason: align with simulator telemetry; preserve "no future values".)*
- **Model packaging:** Exported the trained Keras weights and **standardization/encoding artefacts** for deterministic inference; added a small wrapper to load artefacts at agent start and to expose a `predict(features_json)` endpoint over ZMQ. *(Reason: productionize for the demo run.)*
- **Explainability scope:** Kept SHAP offline (optional) for debugging, not in the real-time loop. *(Reason: compute cost at runtime; Farzi notes SHAP for ANN is heavy.)*
- **Environment:** We retained Farzi's pinned library versions for any re-training, but **inference** runs in our project's Python env (no Kaggle dependency). *(Reason: portability to the agent stack.)*

The justification for the change are as below:

1. **Latency & causality:** A control loop needs fresh state → decision → actuation; ZMQ sockets eliminate file-polling delays inherent to CSV pipelines.
2. **Telemetry-driven features:** Computing lags/splines/filters in-agent ensures strict "no future data" while matching Simu5G's live signals.
3. **Operational robustness:** Packaging the model + scalers avoids training-time environment coupling (Kaggle) while keeping Farzi's reproducibility for offline runs.
4. **Clarity & safety:** Keeping SHAP offline preserves demo performance while still enabling traceability during pre-runs.

6.c Mapping to the demo proposal

This section maps the demo's **functional/performance requirements**, **datasets**, **toolsets**, and **test cases** to the proposal's clauses:

- **Architectural baseline:** Closed-loop four-agent system (Prediction → Optimization → Feedback → Orchestration) with **Simu5G** as a high-fidelity digital twin; **ZeroMQ** is the IPC between Python agents and the C++ simulation.
- **Energy model:** Custom **state-based gNB energy model** (active / micro-sleep / deep-sleep) with active power coupled to **PRB utilization**.
- **Demo flow & success signal:** Observe state → predict → optimize → validate → actuate; show **lower gNB power** with **QoS within policy** and compare to a **baseline** run.

Functional requirements (mapped)

Req-1 (critical): Low-latency simulator–agent integration.
A socketed interface (**ZeroMQ/TCP**) must be integrated between Simu5G and the agents, so decisions are based on fresh state (file polling is not acceptable). **Rationale:** predictive closed loop requires tight latency/jitter bounds.

Req-2 (expected): Policy-guarded actions.
The **Feedback Agent** must validate every proposed plan against 3GPP-aligned throughput/latency guard-rails before actuation; unsafe plans are revised or rejected.

Req-3 (added value): Non-disruptive orchestration.
In-simulation **Orchestration Agent** applies TX-power trims, sleep-state transitions, or carrier on/off **without destabilizing UE KPIs**, with automatic rollback if guard-rails are breached.

Performance requirements:

- Demonstrate **≥15–25% energy reduction** on targeted gNBs **with QoS within policy** (≤3% p95 throughput degradation; latency/loss within limits), compared against an unoptimized baseline.
- Log vectors/scalars to evidence **power draw decrease** and **stable throughput** and present a **baseline vs. optimized** comparison.

Datasets used:

- **Training (offline):** The three **Farzi** CSV datasets — **BS info, Cell-level (hourly), Energy consumption**.
- **Runtime (online):** Live **Simu5G telemetry** (PRB utilization, active UEs, throughput/latency) streamed to the Prediction Agent.

Toolsets:

- **Simu5G (OMNeT++/INET)** as the digital twin with extended **INET Energy Framework** for the gNB power model.
- **ZeroMQ** libraries for C++ (simulation) and Python (agents) to ensure **brokerless, asynchronous IPC**.
- **Python (Keras)** for the **ANN predictor** adopted from Team Farzi; **SHAP** for offline explainability.

Test cases corresponding to the requirements:

TST-1 — Closed-loop energy saving
Trigger: Configure a low-load window; agents run the loop.
Expected result: Action (e.g., "gNB-k → micro_sleep" or "ΔP = −2 dB") is executed; **power**

vectors decrease while **UE throughput remains within policy**. Evidence via Simu5G vectors/scalars and before/after comparison.

TST-2 — Intent-to-policy orchestration
Trigger: Operator intent: "minimize energy with p95 throughput ≥ X and latency ≤ Y."
Expected result: Intent is translated to policies; when conditions are satisfied, an energy action is applied; an **explanation trace** and policy checks are logged.

TST-3 — Safety & rollback
Trigger: After an action, KPIs breach guard-rails.
Expected result: **Feedback Agent auto-reverts** to last safe configuration; event is recorded; KPIs stabilize.

TST-4 — IPC robustness
Trigger: Increase agent–sim exchange rate; introduce jitter.
Expected result: **ZeroMQ** channel sustains low-latency exchanges without stale state; closed-loop remains stable (no oscillations).

TST-5 — Policy conformance under stress
Trigger: Spike traffic to push PRB%; propose aggressive power trim.
Expected result: **Feedback** rejects or down-scales action; **QoS guard-rails** maintained.

TST-6 — Baseline vs. optimized report
Trigger: Run baseline (no control) and optimized (closed loop) with identical seeds.
Expected result: Report shows **≥15–25% energy reduction** with QoS within limits; include vectors/scalars and KPI tables.